Measuring the Carbon Footprint of Programming Languages

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Abstract

This project investigates how programming language choice affects the carbon footprint produced during a program execution. Although software is often seen as intangible, every program consumes energy, ultimately contributing to carbon emissions. As the digital system continues to grow, understanding the environmental impact of software has become increasingly important. This experiment studies how identical algorithms behave when implemented in different programming languages running on the same hardware. By measuring and estimating the carbon emissions, the project tends to explore whether programming language designs play an important role in software sustainability, or if it is mostly driven by heavy algorithms. This research combines the aspects of computer science and environmental science to better understand the hidden environmental cost of everyday computational tasks and encourage more sustainable and responsible software development practices.

Keywords: carbon footprint, software sustainability, energy consumption, carbon intensity, programming languages, algorithms

1 Introduction

As the tech continues to evolve, software becomes an increaingly central part of our lives. From the apps on our phones to the software running our cars, it's everywhere. Everytime a piece of code runs, it triggers energy usage inside the CPU, GPU, and other hardware components [1]. This subsequent usage of software adds up to the carbon emission a device generates [2]. The carbon footprint of software is a growing concern, as it contributes to climate change and has a significant impact on our planet.

The scale of this concern is larger than most people realize. The information and communication technology (ICT) sector accounts for an estimated of 2.1% to 3.9% of the global greenhouse gas emissions [3]. The global data center energy consumption alone is projected to reach around 945TWh by 2030, which is a significant increase from 153TWh in 2005, 205TWh in 2018. [4, 5]. Unlike the visible emissions from vehicles, software carbon is invisible. There is no smoke when a program runs, yet every computation contributes indirectly to the same atmosphere. This invisibility is precisely what makes it dangerous. It does not trigger the same urgency that the physical pollution does, even as the number grows.

One of the most consequential and least-studied of those decisions is the choice of programming language. The carbon cost of a running program is not determined by a hardwar alone. It is also shaped by how efficient the language runtime executes instructions, manages memory, and utilizes the CPU.In a landmark empirical study, Pereira et al. [6] benchmarked 27 programming languages across 10 problems drawn from the Computer Language Benchmarks Game [7], measuring energy consumption via Intel's Running Average Power Limit (RAPL) interface. They found that the choice of programming language can have a significant impact on energy consumption, with some languages consuming up to 2.5 times more energy than others for the same task. For example, they found that C and Rust were among the most energy-efficient languages, while Python and JavaScript were among the least efficient. Python consumed approximately 75.88 times more energy per benchmark than C while JavaScript consumed approximately 4.45 times more energy per benchmark than C [6]. Since carbon emissions are directly proportional to energy consumption, these differences are not just technical concern, but also a carbon concern. A Python program consuming about 75 times more energy than an equivalent C program does not just run slower, it emits 75 times more carbon for every execution. Now, multiply that by the millions of times a piece of software runs, and the carbon cost becomes staggering.

Prior work in this area, including Pereira et al., reports energy consumption in joules or kilowatt-hours but none of it completes the translation into carbon footprint. This

is a critical gap. Energy consumption is the underlying driver, but carbon footprint is the ultimate impact. This unit is most directly relevent to the clomate policy and environmental decision making. Henderson et al. showed that the emissions can be reduced by up to 30 times just by running the experiments in the locations powered by more renewable energy sources [8].

This paper presents a controlled empirical study comparing the energy consumption and the carbon footprint of six programming languages (C, Rust, Go, Java, Javascript, and Python) across six different algorithms and three imput sizes. The energy consumption is measured using Intel's RAPL (accessed through LibreHardwareMonitor [9]), and real-time carbon intensity data is fetched from the Electricity Maps API [10] for the New York grid zone (US-NY-NYIS). Each of the 108 experimental cells (6 languages $\times$ 6 algorithms $\times$ 3 input sizes) was replicated 115 times, yielding 12,420 total measurements. Statistical analysis applies Welch's t-tests with Bonferroni correction across all 15 pairwise language comparisons. The results show that 14 of 15 language pairs are statistically distinguishable in carbon emissions. The sole exception is C versus Rust ($t = -0.313$, $p = 0.754$), confirming that the two languages are carbon-equivalent. Python is the most carbon-intensive language tested, producing on average 213$\times$ more carbon emissions than C for identical workloads.

2 Background

In the following sections, we describe the energy consumption model, carbon intensity assumptions, programming languages, and the implementations of the six algorithms used in this study.

2.1 Energy Consumption in Software

To quantify the energy consumed by a software program, we adopt the standard defination of energy as the capacity to do work.

$$E_{\text{energy}} = P_{\text{power}} \times t_{\text{time}} \tag{1}$$

where E is energy measured in Joules(J), Power is rate of energy consumption measured in Watts(W), and Time is the duration of execution measured in seconds(s)[11]. In the context of software execution, the T refers to the wall time, the elapsed time from the first instruction of the measured code path to the last, including CPU computation, memory access, and all other processor activity during execution [12].

The energy consumed by a CPU package during the program execution can therefore be decomposed as the integral of instantaneos power draw over the execution window. Since power draw is not constant across languages or algorithms, a key finding of Pereira et al. [6], energy efficiency cannot be inferred from execution time alone. A faster program is not necessarily a more energy efficient one. This encourages us to direct hardware level energy measurement via RAPL.

All experiments were conducted on a desktop with the following specifications: Windows 11 Enterprise 25H2 (Build 26200.8246) operating system, with 32.0 GB RAM, and an Intel Core `i7-14700K` CPU @ 3.40 GHz.

2.2 Carbon Intensity

To translate the energy measurements into carbon emissions, we apply the SCI framework.

The relationship between energy and carbon depends on how that energy is generated. Energy produced from hydroelectric or wind generation carries near-zero carbon intensity while the electricity from coal or natural gas has 800-1000 gCO2/kWh [13]. The carbon intensity of an electricity grid represents the average emissions intensity of all generation sources dispatched at a given moment, weighted by their contribution to total supply. This intensity is measured in grams of CO2 equivalent per kilowatt-hour, gCO2/kWh.

We obtain real-time carbon intensity from the Electricity Map API [10], which provides grid-zone-level carbon intensity updates every 30 minutes. Our experiment used zone US-NY-NYIS (New York Independent System Operator), which had a carbon intensity of `300-306` gCO2/kWh during our overnight experiment window (10pm - 6am). The gCO2 metric is computer from RAPL PKG energy readings as follows.

$$\text{gCO}_2\text{e} = \frac{E_{\text{J}}}{3{,}600{,}000} \times I_{\text{carbon intensity (gCO2/kWh)}} \tag{2}$$

where I is the grid carbon intensity in gCO2/kWh. This methodology is consistent with the Green Algorithms framework [14] and the Software Carbon Intensity (SCI) specification [15]

2.3 Programming Language

The programming languages chosen for this study are C, Python, Javascript, Java, Go, and Rust. These languages were selected to represent a range of programming paradigms, performance characteristics, and popularity in the software development community. Each language has its own unique features and trade-offs that may influence the energy consumption and carbon footprint of the algorithms implemented in them [16]. The performance characteristics of these languages are summarized in Table 1.

Table 1: Programming languages, their performance characteristics, and popularity.

Language	Paradigm	Runtime	Version
C	Manual Memory, Compiled	GCC	14.3.0
Rust	Ownership, Compiled	Rustup	1.94.0
Python	Interpreted	CPython	3.12.10
Javascript	Dynamic, V8	Node.js	24.14.0
Java	OOP, JVM	OpenJDK	21.0.10
Go	GC	Go	1.26.1

GCC was installed via WinLibs (MingGW-w64 distribu-

tion for Windows). Rust was installed via Rustup. Python was installed via the official Python installer for Windows. Java was installed via the official OpenJDK distribution for Windows. Go was installed via the official Go installer for Windows. Node.js was installed via the official Node.js installer for Windows.

C provides direct hardware access with no runtime overhead and serves as the energy baseline for all normalized comparisons.

Rust acheives memory safety without a garbage collector through a compile-time ownership and borrowing checking system. It delivers performance comparable to C while preventing common memory errors, which may lead to more efficient energy usage in certain workloads.

Go compiles to native machine code but includes a concurrent garbage collector that runs alongside the program, introducing periodic overhead that is captured within the RAPL measurement window.

Java executes on the Java Virtual Machine (JVM) and uses Just-In-Time (JIT) compilation to optimize code at runtime. The JVM's garbage collector can introduce additional overhead, and the performance can vary based on how well the JIT optimizes the code during execution.

JavaScript uses a JIT compiler with dynamic typing and a garbage collector, which can lead to variable performance and energy consumption depending on the workload and how well the JIT optimizes the code.

Python is a bytecode-interpreted language in which every operation passes through a C-level interpreter dispatch loop. It has a global interpreter lock (GIL) that limits concurrent execution of threads, and its dynamic typing and high-level abstractions can lead to increased energy consumption compared to compiled languages.

2.4 Algorithms

The algorithms were chosen to span a range of time complexities and computational patterns, including CPU-bound, memory-bound, and branching-heavy workloads. The six algorithms used in this study are summarized in Table 2.

Table 2: Algorithms, their time complexities and computational patterns.

Algorithm	Complexity	Pattern
Summation	$O(n)$	Linear
Binary Search	$O(\log n)$	Logarithmic
Merge Sort	$O(n \log n)$	Divide & Conquer
BFS	$O(V + E)$	Graph Traversal
Hash Table	$O(n) - avg$	Memory Bound
Matrix Multiplication	$O(n^3)$	Dense Arithemetic

The pseudocode for each algorithm, as implemented indentically across all languages, is provided below. The implementations are designed to be as similar as possible across languages to ensure a fair comparison of their energy consumption and carbon footprint.

Summation - This algorithm performs a single sequential sum over an array of integers. It is a simple CPU-bound with no branching and minimal memory access, making it a baseliune workload that reflects the idle runtime cost of the language.

Binary Search - This algorithm halves the search interval over a sorted array. Its time complexity is logarithmic, and it involves branching based on comparisons, which means very few operations are performed at any input size. So, the observed energy difference between languages may be more influenced by the runtime efficiency and overhead than the algorithmic complexity.

Breadth-First Search - This algorithm traverses a graph level by level from a source node. It is memory-bound. The primary cost is queue operations and random memory access patterns that can lead to cache misses. The energy consumption may be more sensitive to how efficiently the language runtime manages memory and data structures.

Merge Sort - This algorithm recursively divides the array into halves and merges them back in sorted order. Now, just to make sure that we don't get any RecursionError in Python, we will increase the recursion limit for merge sort.

Hash Table - This algorithm use open addressing with linear probing. Each operation is O(1) on average, but can be O(n) in the worst case if all keys collide. The energy consumption may be influenced by how well the language runtime optimizes for hash table operations and memory management. the "& 0xFFFFFFFFFFFFFFFF" is used to ensure that the hash result is always a 64bit integer, which is important for consistency across different platforms and languages. This way, we can compare the hash results from C and Python without worrying about differences in integer sizes or overflow behavior.

Matrix Multiplication - This algorithm uses the naive triple loop method to multiply two square matrices. It is a dense arithmetic workload with cubic time complexity, which means the number of operations grows rapidly with input size. At n=500 (our large input), it performs 125 million multiplications and additions, which is a significant computational load that can highlight differences in how languages handle CPU-bound tasks and memory access patterns.

3 Methodology

This section describes the experimental setup, the measurement protocol used to isolate algorithm energy consumption, the implementation constraints across all six languages, the input sizes used for each algorithm, and idk.

3.1 Experimental Setup

All 12,420 experiments were conducted on a single dedicated machine to eliminate any inter-machine power variability. The Intel's RAPL interface exposes hardware energy counters directly to software, enabling fine-grained CPU package energy measurement without externam instruments [17].

3.1.1 Implementation-specific design decisions

All 36 implementations (6 languages x 6 algorithms) were written from scratch following a shared pseudocode specification. The following principles were followed to ensure a fair comparison across languages:

Algorithmic consistency - All implementations strictly follow the same algorithmic logic and data structures as defined in the pseudocode specification. No language-specific optimizations or library functions that could alter the algorithm's behavior are used. Every algorithm is implemented from primitive operations only like loops, conditionals, integer arithmetic and array indexing.

Reset before each measurement - Data structures that are mutated during algorithm execution are reset to their initial state at the beginning of each algorithm invocation. Critically, this reset happens before the energy measurement window opens, so the reset cost is not included in the measured energy. This guarantees that every replicated run measures the same algorithmic work.

Output verification - Each implementation produces a checksum output that is verified against a reference implementation to ensure correctness. This ensures that all implementations are performing the same computations and producing the same results.

Input size consistency - The same input sizes are used for each algorithm across all languages to ensure that we are measuring comparable workloads. This allows us to analyze how energy consumption scales with input size for each algorithm and language.

Single checksum output - Each implementation produces a single checksum output, and prints it to stdout after each algorithm execution. This is used for cross-language output verification and does not affect the measured energy as it occurs inside the measurement window but is identical and negligible across all languages.

3.2 Stdin Measurement Protocol

The main challenge in cross-language energy measurement is to ensure that we are measuring the energy consumed by the algorithm itself, and not the overhead of the language runtime or other background process. To address this, we designed a custom measurement harness that follows a strict protocol to isolate the algorithm execution. The harness uses a stdin/stout handshake design in which the program initializes once and then loops, executing the algorithm on each trigger without restarting the process.

The measurement sequence is shown in the Listing 1.

```
setup - not timed
print "ready" - not timed
harness newline - start timer
run command - measurement
print checksum - stop timer
flush output - not timed
```

Listing 1: Measurement sequence

The flush() call on line 7 is critical. Without it, the newline sits in a kernel pipe buffer and the program never unblocks, causing the measurement to hang indefinitely. On the program side, each language blocks on its respective stdin/stdout primitive until the trigger arrives, executes the algorithm, and writes the checksum. Listing 2 shows the blocking call used in each language.

```
// C
while (fgets(buf, sizeof(buf), stdin))  {
    run()}

// Python
for line in sys.stdin: run()

// JavaScript
rl.on('line', () => run())

// Java
while(br.readLine() != null) {run()}

// Go
for scanner.Scan() {run()}

// Rust
for line in stdin.lock().lines() {run()}
```

Listing 2: Stdin primitives in each language implementation

C uses fgets rather than scanf because it reads exactly one line including the newline, giving unambiguous trigger semantics. Java uses BufferedReader.readLine() rather than Scanner to avoid locale-aware parsing overhead inside the measurement window. JavaScript uses Node.js's readline event listener. Python uses for line in sys.stdin: rather than repeated input() calls to avoid Python-level function call overhead per iteration.

3.3 Experimental Execution

Warm-up Runs - Five runs per cell are discarded before measurement was recorded to allow the JIT compilers and

garbage collectors in languages like Java and Go to warm up and reach a steady state. For C and Rust, warm-up primarily ensures that the instruction cache and branch predictor are in a warmed state consistent with repeated execution.

Cell Sleep - A mandatory 5s sleep is inserted between cells to allow the processor's thermal management to return to stable baseline between cells. This reduces the rist that thermal throttling in one cell biases the initial temperature state of the next.

Experiment timing - All 36 language-algorithm-size combinations are executed in a single overnight run from 10pm to 6am to minimize background system load and to capture a consistent grid carbon intensity profile during the measurement window.

Carbon Intensity Refresh - The Electricity Maps API queried once at experiment start and then refreshed every 30 minutes. Two carbon intensity values were returned during the experiment window i.e. 300 gCO2/kWh and 306 gCO2/kWh, representing a 2% spread in the carbon intensity during the experiment window.

3.4 Output Verification

Before energy measurement, all 36 implementations are verified for correctness. The verification script compiles and runs each implementation at all three input sizes, compares the checksum printed to stdout against the reference checksum produced by the C implementation, and assigns each language-algortihm-size combination a PASS/FAIL status. Table 3 shows the verification results fot the C implemetation. The C times serves as a reference baseline against which all other languages are measured and normalized. They confirm that execition time scales with the expected time complexity of each program.

The execution times in Table 3 confirms the time complexity scaling. Merge sort grows from 0.0428ms to 24.740ms (a 576× increase) as input scales from small to large with consistent $O(n \log n)$ growth.

3.5 Input Sizes

Three input sizes are used for each algorithm to capture scaling behavior: small, mid, and large. The specific sizes are chosen to be large enough to produce measurable energy consumption while still completing within a reasonable time frame across all languages. The input sizes for each algorithm are as defined in Table 4.

3.6 Pilot Study and Sample Size Determination

Before the main experiment, a pilot study of 50 iterations per cell was conducted to esitmate per-cell variance and determine the required sample size [18]. Per-cell coefficients of variation can be calculated with the following formula:

$$CV = \frac{\sigma}{\mu} \times 100\% \tag{3}$$

Table 3: C — RUN & VERIFY

Algorithm	Size	Output	Time	Status
`summation`	small	5050	0.0437ms	PASS
`summation`	mid	50005000	0.0406ms	PASS
`summation`	large	500000500000	0.4715ms	PASS
`binary_search`	small	99	0.0448ms	PASS
`binary_search`	mid	9999	0.0392ms	PASS
`binary_search`	large	999999	0.0835ms	PASS
`merge_sort`	small	100	0.0428ms	PASS
`merge_sort`	mid	10000	0.2459ms	PASS
`merge_sort`	large	1000000	24.7405ms	PASS
`bfs`	small	100	0.0430ms	PASS
`bfs`	mid	1000	0.3039ms	PASS
`bfs`	large	5000	7.2774ms	PASS
`hash_table`	small	64850	0.0433ms	PASS
`hash_table`	mid	649985000	0.2467ms	PASS
`hash_table`	large	64999850000	2.1579ms	PASS
`matrix_mul`	small	9175	0.0744ms	PASS
`matrix_mul`	mid	646700	2.5604ms	PASS
`matrix_mul`	large	10291750	47.1134ms	PASS

Table 4: Input Sizes per Algorithm

Algorithm	Small	Mid	Large
`summation`	100	10,000	1,000,000
`binary_search`	100	10,000	1,000,000
`merge_sort`	100	10,000	1,000,000
`bfs`	100	1,000	5,000
`hash_table`	100	10,000	100,000
`matrix_mul`	50	200	500

Where σ is the standard deviation of the energy measurements for a given cell, and μ is the mean energy measurement for that cell. The results revealed a wide and informative range. CV spanned from 4.4% (Python matrix-multiplication, mid) to 122.7% (JavaScript merge-sort, small), with an overall mean CV of 38.4% and median of 43.0%. The high CV values observed at small input sizes across all langauges are consistent with a known limitation of RAPL-based measurement. When an algorithm completes in microseconds, and OS scheduling event, cache miss, or background interrupt becomes a large fraction of the total measured energy, inflating variance [19]. This is clearly visible in our pilot data. The noisiest cells are uniformly those with short runtimes, while the most stable cells (Python matrix-multiplication, mid: CV=4.4%, large: CV =6.9% and Java merge-sort, large: CV=5.9%) are those where the algorithm's own energy consumption is large enought to dominate the background noise, This pattern directly motivates the focus on large input size results in the main statistical analysis, where RAPL measurements are most reliable.

A formal power analysis was then conducted using the `TTestIndPower` solver from the Python `statsmodels` library [20] to determine the minimum sample size N per cell required to detect a medium effect size (Cohen's $d = 0.5$) with 80% power at the Bonferroni-corrected significance level of $\alpha' = 0.05/15 = 0.003\overline{3}$, using a two-sided Welch's t-

test. The solver returned a raw N of 116.28, which rounds up to a recommended minimum of $N = 117$. The main experiment used $N = 115$ per cell, 2 runs below the recommended minimum, a 1.7% shortfall. Given the large effect sizes observed between paradigmatically different languages (particularly Python versus compiled languages), this marginal reduction in statistical power is negligible in practice and does not affect any of the study's conclusions. All 108 cells were replicated exactly 115 times, yielding 12,420 total measurements.

baseline power draw - 40w 5s sample

4 Statistical Analysis

4.1 Shapiro-Wilk

For each cell, it tests the null hypothesis that the data is normally distributed. It returns two value.

5 Results

Before presenting results, we report the integrity of the collected data. The benchmark runner produced a CSV file containing 12,420 rows, exactly 115 runs for each of the 108 cells. All 108 cells returned a consistent checksum across all 115 runs, confirming that ever implementation produced identical output on every invocation. The carbon intensity column contained exactly two distinct value

6 Discussion

interesting observation. I has a youtube video playing and it was boosting my joules. once I stopped it, it went to baout 300 from 500/600.

7 Future Work

Cross-Architecture Measurement - This study was exclusively conducted on an Intel Core i7-14700K CPU, which is a high-end desktop processor. Now, an open question is if the language hierarchy identifies here generalizes to other architectures. Processors like ARM, Ryzer, and Apple M-series etc. dominate the mobile and laptop market now, and differ from x86 architecture in terms of energy cost, branch prediction, and memory management. Future work can extend this experiment to a more diverse set of hardware platforms to see if the language hierarchy holds across different architectures. Additionally, the experiment can be repeated on a more recent generation of CPUs to see if the language hierarchy changes with newer hardware optimizations and energy efficiency improvements.

Cloud Computing Energy Measurement - All measurements in this study were collected on a local desktop machine where Intel RAPL is directly accessible via a kernel driver. Extending this methodology to public cloud environments like Amazon EC2, Google Cloud, or Microsoft Azure is both a challenge and an opportunity for future work. Cloud providers typically do not expose hardware-level energy measurements to users, and the virtualized nature of cloud instances adds complexity to isolating the energy consumption of specific workloads. Future research can explore alternative approaches for estimating energy consumption in cloud environments, such as modeling based on resources used (CPU time, memory usage etc) and the known energy profiles of the underlying hardware.

8 Conclusion

Acknowledgements

This project made a limited use of AI toolsduring the software development phase, specifically for generating initial code snippets and providing suggestions for code improvements. No AI was used during the statistical analysis, or the writing of this paper itself. In the spirit of this paper's subject matter, the author acknowledges the energy cost of the research itself. The main benchmark experiment recorded approximately 283,654 J of CPU package energy across all 12,420 runs. An estimated additional 50,000 J of untracked energy was consumed during warm-up runs, verification runs, and development and debugging of the measurement infrastructure.

measure later

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